

CHAPTER 7

Factors

Chapter Summary

Factors drive risk premiums. One set of factors describes fundamental, economy-wide variables like growth, inflation, volatility, productivity, and demographic risk. Another set consists of tradeable investment styles like the market portfolio, value-growth investing, and momentum investing. The economic theory behind factors can be either rational, where the factors have high returns over the long run to compensate for their low returns during bad times, or behavioral, where factor risk premiums result from the behavior of agents that is not arbitrated away.

1. Value Investing

Historically speaking, value stocks beat the pants off growth stocks. Value stocks have low prices in relation to their net worth, which can be measured by accounting book value. Growth stocks are relatively costly in comparison to book value. Figure 7.1 plots the returns of value stocks (stocks with high book-to-market ratios) versus growth stocks (stocks with low book-to-market ratios). I plot the returns to the *value-growth strategy*, which goes long value stocks and short growth stocks.¹ Although value investing has on average done well, it sometimes loses money. For example, note the pronounced drawdown during the tech boom of the late 1990s. There was another drawdown during the financial crisis in 2008. Value stocks also did poorly in 2011.

Why does value investing work? Was the value strategy—the returns of value stocks in excess of growth stocks—a systematic factor? If so, what determined the value risk premium?

In the context of the previous chapter on factor theory, assets are buffeted by risk factors. The risk factors offer premiums to compensate investors for bearing

¹ The data for this strategy, as for all the other Fama–French strategies in this chapter are from http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.